

ANIK DUTTA

Full-Stack Engineer | Generative AI Developer | Cloud & MERN Stack Specialist

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Goal-oriented Software Engineer, targeting **Full-Stack Development, Cloud Engineering, and Generative AI** roles to build scalable, intelligent software solutions that drive innovation, optimize operations, and create measurable business impact.

PROFILE SUMMARY

- **Software Engineer** with 2+ years of experience in **Full-Stack Development, Cloud Infrastructure, and Generative AI**, with expertise in **MERN stack, Next.js, TypeScript, and Python** for building scalable, high-performance applications.
- Skilled in developing **AI-driven applications and intelligent automation systems** using OpenAI, Google Gemini, RAG pipelines, vector databases, ElevenLabs, LangChain, and the Claude Model Context Protocol (MCP).
- Experienced in **cloud-native architecture and DevOps**, leveraging AWS, Google Cloud, BigQuery, Kubernetes, and CI/CD practices to ensure scalable and reliable production systems.
- Proven team leader currently **leading a 3-member engineering team**; migrated **500,000+ user records with zero data loss** during a large-scale platform modernization.

CORE COMPETENCIES

- ➔ Full-Stack Web Development (MERN Stack)
- ➔ Cloud Infrastructure & Deployment (AWS / GCP)
- ➔ DevOps & CI/CD Automation
- ➔ AI & Generative AI Application Development
- ➔ Microservices & Scalable System Architecture
- ➔ REST API Design & Integration
- ➔ Database Design & Data Pipelines
- ➔ Front-end Engineering (React / Next.js)
- ➔ Back-end & API Development
- ➔ Agile Development & Team Leadership
- ➔ Performance Optimization
- ➔ Data Structures & Algorithms (C++)

TECHNICAL SKILLS

- ❖ **Programming Languages:** JavaScript, TypeScript, Python, C++, SQL
- ❖ **Web Development:** MERN Stack, Next.js, React Native, Redux, Tailwind CSS
- ❖ **Back-end:** Node.js, Express.js, NestJS, Flask, REST APIs, GraphQL
- ❖ **Cloud Platforms:** AWS, Google Cloud, BigQuery
- ❖ **DevOps & Deployment:** Docker, Kubernetes, Terraform, Git, GitHub, CI/CD, SFTP, Linux
- ❖ **Databases:** MongoDB, MySQL, PostgreSQL, Oracle, Redis, Vector DB (Pinecone / Chroma)
- ❖ **AI & Generative AI:** OpenAI, Google Gemini, Vertex AI, LangChain, RAG, Claude MCP, ElevenLabs

EDUCATION

- **2024:** B.Tech in Computer Science & Engineering, Sister Nivedita University, Newtown, WB
- **2020:** Higher Secondary (Class XII), Hooghly Branch Govt School

SOFT SKILLS



- Stakeholder Collaboration
- Analytical Thinking
- Problem Solving
- Effective Communication
- Leadership & Team Facilitation
- Adaptability & Agile Mindset

WORK EXPERIENCE

Aug 2024 – Present: Software Engineer | Aeonix Research & Innovations LLP — Kolkata, India

Key Result Area:

- Leading a **3-member engineering team**, driving architecture decisions, code reviews, sprint planning, and delivery of scalable full-stack products.
- Designing and building an **Elder Care Management System** using **Next.js, Node.js (Express), TypeScript, and MongoDB** with secure, high-throughput backend services.
- Provisioned and managed cloud infrastructure as code using **Terraform**, and implemented **Redis** caching to accelerate API response times and reduce database load on the Elder Care platform.
- Developing a mobile/web application integrated with an **SOS device** for real-time alerts and data synchronisation.

- ▶ Integrated **Google BigQuery** with automated **daily data syncs** from MongoDB for centralized analytics, reporting, and data warehousing.
- ▶ Building **Generative AI** features and AI agents using OpenAI, Google Gemini, RAG, vector databases, ElevenLabs, and Claude MCP.

Key Highlights:

- ▶ Migrated **500,000+ legacy user records** to the new platform with **zero data loss**.
- ▶ Optimized Next.js/Node.js components via server-side rendering, caching, and memoization to significantly improve application performance.

INTERNSHIP

Aug 2023 – Aug 2024: Software Engineer Intern | Aeonix Research & Innovations LLP — Kolkata, India

Key Result Area:

- ▶ Built and launched a **Group Insurance Enrollment Portal** for the US market using **React, Node.js, Express, and MongoDB**.
- ▶ Established a secure data transmission channel to **21 insurance companies**, fetching 10,000+ records from MongoDB and transmitting via **SFTP** weekly.
- ▶ Oversaw onboarding of 21 insurance company policies — building policy forms in React, enrollment in Node.js, and storage in MongoDB.

Key Highlights:

- ▶ Achieved a **40% increase in UI performance** via Context API, lazy loading, and memoization.
- ▶ Improved database query performance by **70%** through optimized Mongoose queries.

2022 – 2023: Software Developer Intern | Computer Software Solutions LLC (CSSLC) — US-Based Startup

Key Result Area:

- ▶ Developed features for a **US-based job search portal** as a full-stack developer intern, building responsive front-end interfaces and REST APIs.
- ▶ Implemented job listing, search, and application workflows using the **MERN stack** (MongoDB, Express.js, React.js, Node.js).
- ▶ Collaborated with a distributed team in an Agile environment to ship and iterate on product features.

Key Highlights:

- ▶ Contributed to a production job-search platform for a US-based startup, enhancing search functionality and overall user experience.

PROJECTS

- ▶ **Personal Assistant Using Generative AI:** AI-powered assistant built with LangChain and OpenAI APIs for natural-language queries and task automation; integrated Google Calendar API for real-time scheduling, reminders, and conflict detection.
- ▶ **Social Media Automation AI Agent:** AI content pipeline using Veo 3 (Vertex AI) to generate and schedule posts across Facebook, Instagram, and YouTube; used Heygen and Midjourney for creative asset generation.
- ▶ **Real Estate Plot Detection:** Used OpenCV and Python Flask to detect plots from layout images and convert them into SVG format.
- ▶ **Scalable & Resilient Cloud Architecture:** Fault-tolerant AWS system using API Gateway, Lambda (serverless), and EC2 Auto Scaling Groups to sustain performance during high-traffic periods.

CERTIFICATIONS

- ▶ Store, Process, and Manage Data on Google Cloud — Google Cloud
- ▶ Deploy Kubernetes Applications on Google Cloud — Google Cloud
- ▶ JavaScript Developer Certificate — HackerRank

 **Location:** Kolkata, West Bengal, India | **Languages Known:** English, Hindi, Bengali